Chest Wall Deformities

The Big Picture

Four deformities to know: **scoliosis**, **pectus excavatum** (funnel/caved-in), **pectus carinatum** (pigeon/keel), and **Poland syndrome**. Themes: how each looks, how it affects lung/heart function, the **measurement indices** (Cobb's angle, Haller index), and when to operate.

Scoliosis

- **Lateral curvature of the spine + rotation** of vertebrae & viscera; commonest deformity. May (usually doesn't) have kyphosis.
- **Causes:** non-structural (postural, sciatic, hysterical) vs **structural (idiopathic — commonest, congenital, neuropathic — polio/CP/neurofibromatosis, myopathic — Duchenne, hereditary — Marfan/Ehlers-Danlos, Turner)**, plus lung fibrosis/burns/irradiation.
- **Most common:** thoracic curve **to the right**, **girls**, higher angle = worse prognosis.
- **Cobb's angle** — angle between perpendiculars off the end-vertebrae. **0–10° normal; >10° = scoliosis.**
- **Lung effect: restrictive pattern** — ↓ VC, TLC, MVV, FRC, PEF, **DLCO**; FEV1/FVC normal; compliance ↓. Severe **(≥100°)** → sleep desaturation, **pulmonary hypertension** → **RVH/cor pulmonale**.
- **Death** commonly from **respiratory failure** (repeated chest infection) or cardiac disease; PHT/RVH in 4th decade.
- **Treatment** — surgery (correct angle, halt progression) for symptomatic/dyspnoeic; **spinal fusion** (congenital/KS); **growth-friendly rods** (magnetically controlled, titanium rib) in children.

Pectus Excavatum (Funnel Chest)

- **Most common chest deformity** (thoracic surgery); **depressed sternum + adjacent ribs**, caved-in. 1st/2nd ribs + manubrium normal; **right often more depressed**.
- Present at birth/1st year, **worsens in adolescence**; abnormal cartilage growth / collagen defect (**Marfan**); associated scoliosis, asthma.
- **3:1 male**, family history 37–47%; tall, lanky, poor posture.
- **Haller index** — transverse / AP diameter at deepest point. **Normal ~2.5; >3.25 = significant.**
- **Presentation:** usually **cosmetic/asymptomatic**; may have precordial pain, palpitation (**mitral valve prolapse**), systolic ejection murmur, dyspnoea, ↓ exercise tolerance.
- **Surgery** (mostly cosmetic) — best at **10–14 years**. Indications: progressive symptoms, restrictive PFT, **CT cardiac compression**, **Haller >3.25**, atelectasis, MVP/BBB, recurrence.
- **Operations:** silicone implant (cosmetic), **Ravitch** (open — cartilage resection + sternal osteotomy; for combined/asymmetric), **Nuss** (minimally invasive curved metal bar, **left 2–4 yrs** — most common), non-surgical.

Pectus Carinatum (Pigeon Chest)

- **Prominent outward sternum** ("keel of a ship"), ribs falling away.
- **Less common than excavatum**; **M:F 4:1**, family history 26%; **1–3% cardiac anomaly** (coarctation) → **send for echo**.
- Presentation: protrusion ± rib depression, pain, exercise limitation, sternal rotation; chondromanubrial (comma sternum) variant = ↑ heart disease.

- **Treatment: Ravitch**, or **bracing** (compression) in selected/early cases.

Poland Syndrome

- **Absence of pectoralis major (± minor) + ipsilateral syndactyly.**
- → rib hypoplasia on affected side, **hyperlucent lung on CXR**. Cause unknown. Main complaint = **cosmetic**; surgery can separate digits.

Walk-Away Points

- **Scoliosis** = lateral curve + rotation; **Cobb's >10°**; restrictive lung defect, ≥100° → PHT/cor pulmonale; death from respiratory failure.
- **Pectus excavatum** (funnel, commonest) — **Haller index >3.25**, **Nuss** (bar) vs **Ravitch** (open), repair at 10–14 yrs.
- **Pectus carinatum** (pigeon, keel) — echo (cardiac association), Ravitch or bracing.
- **Poland = absent pectoralis + syndactyly**, hyperlucent lung.